WEEK 2 — DATA VISUALIZATIONS

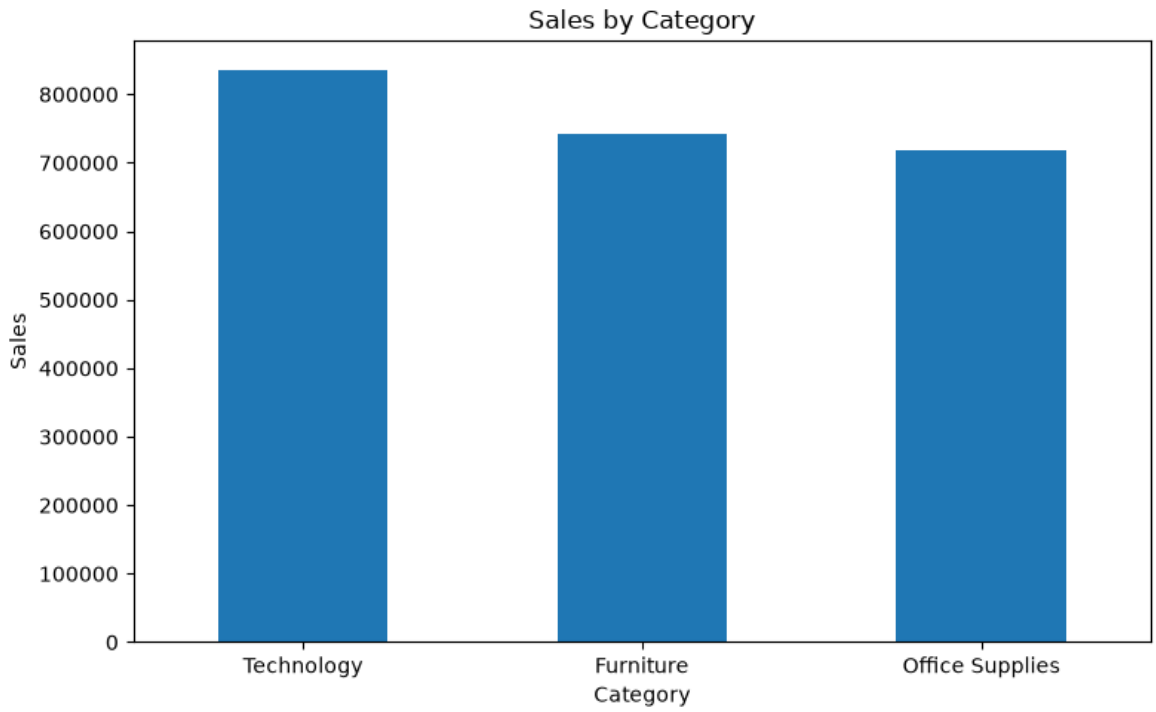
Exploratory Data Analysis & Business Insights — Superstore Dataset

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Project: Data Analysis Internship — NexAfrica
Tools: Python, Pandas, NumPy and Matplotlib

This visualization pack presents the main charts generated during Week 2 Exploratory Data Analysis. The figures are based on the analysis code supplied for the Superstore dataset. They are organised to show overall performance, profitability, discount relationships, trends, regional differences, and product-level loss areas.

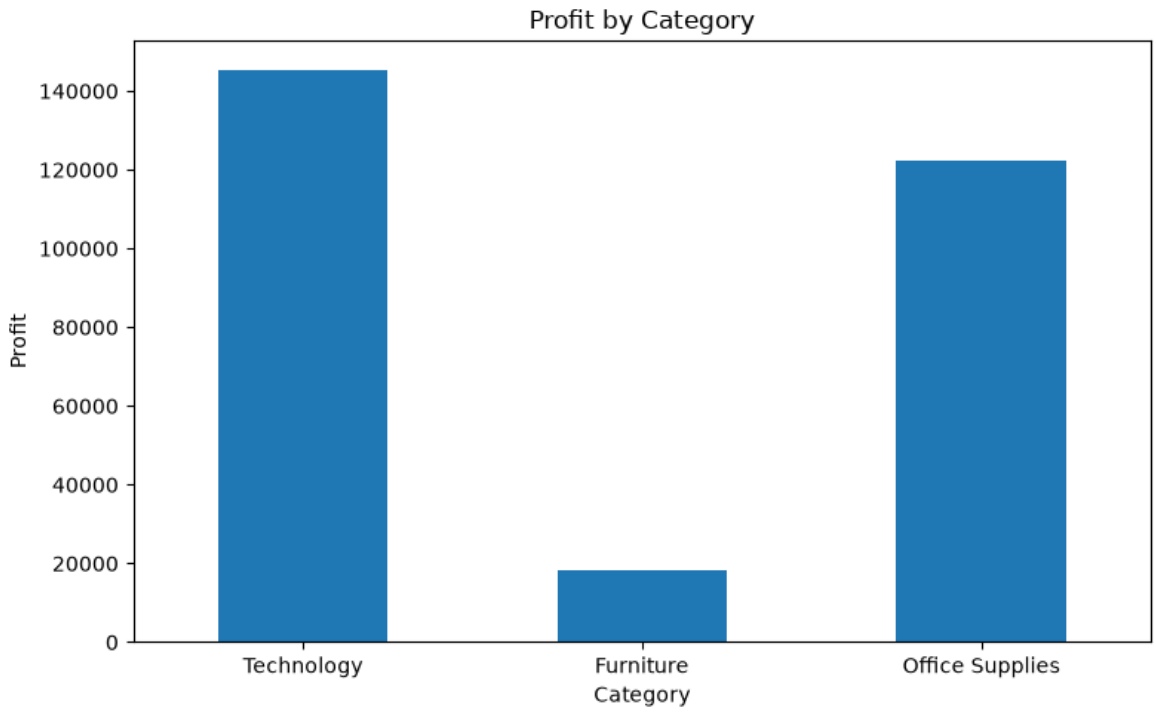
Analysis area	Visualisations included
Category performance	Sales, total profit, profit margin
Discount analysis	Total profit, average profit per record, profit margin
Relationships	Correlation matrix, Sales vs Profit
Trends	Monthly sales trend
Geography	Profit by region
Sub-category / product	Profit by sub-category, top 10 loss-making products

1. Sales by Category



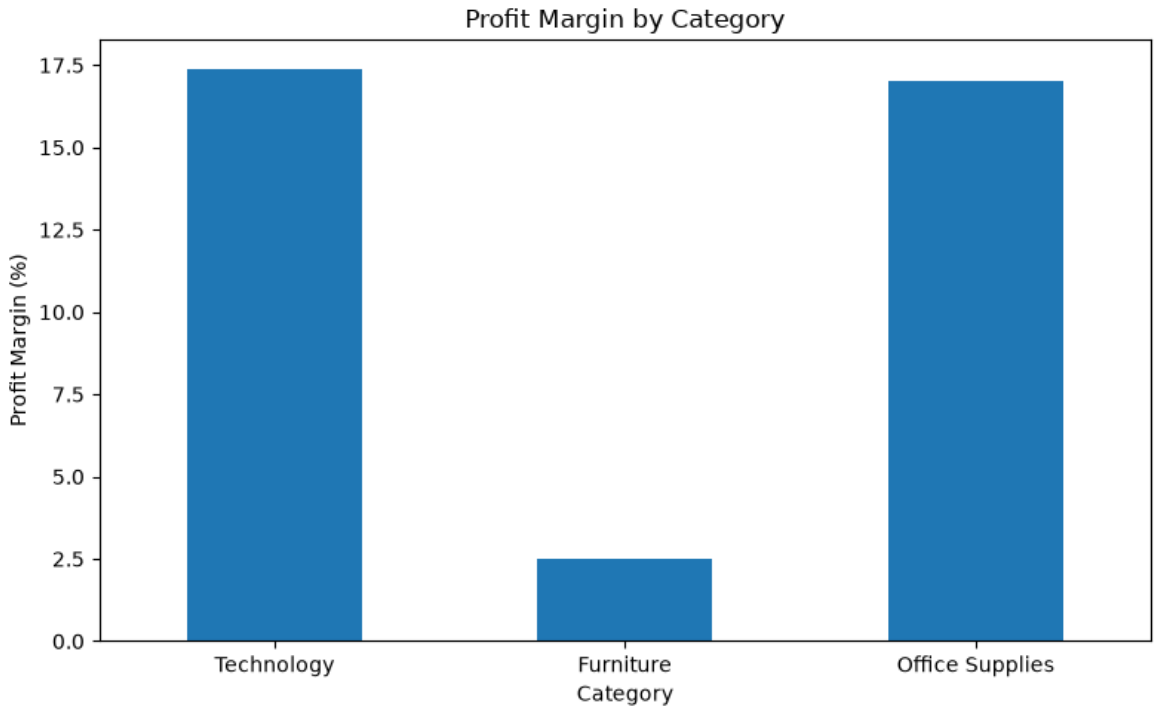
Interpretation: Technology has the highest sales among the three categories in the analysed dataset, followed by Furniture and Office Supplies.

2. Profit by Category



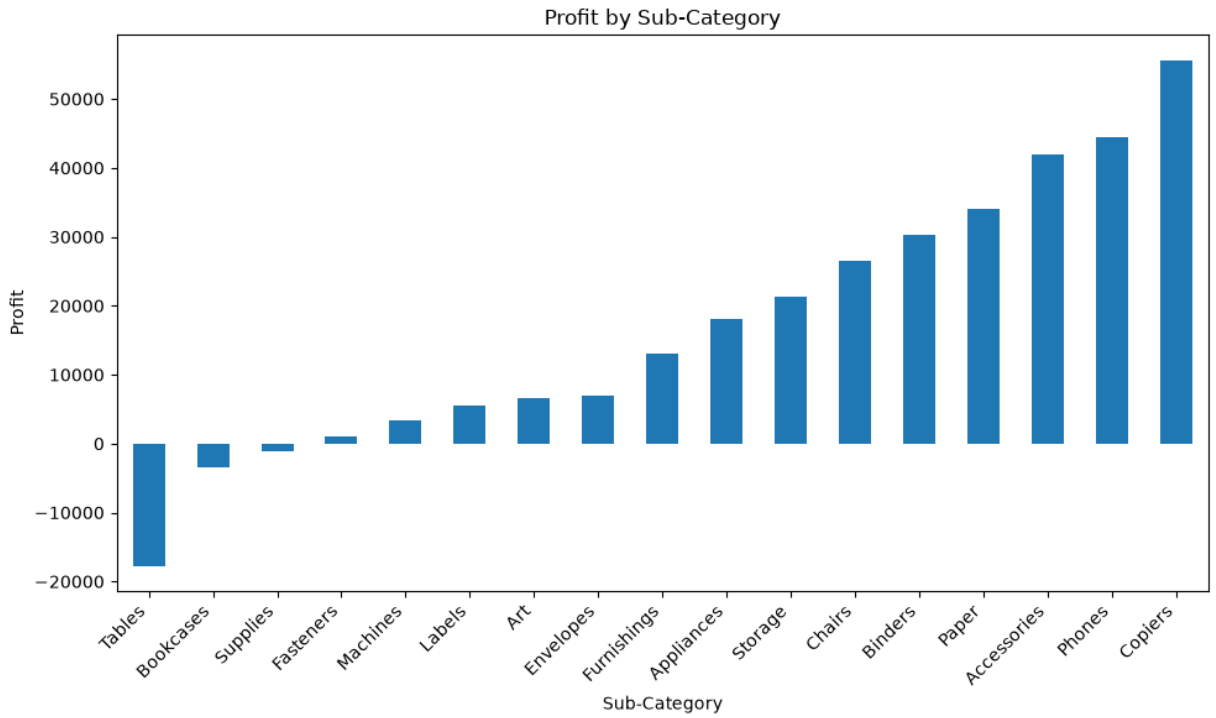
Interpretation: Technology records the highest total profit, while Furniture contributes substantially less profit despite its relatively high sales.

3. Profit Margin by Category



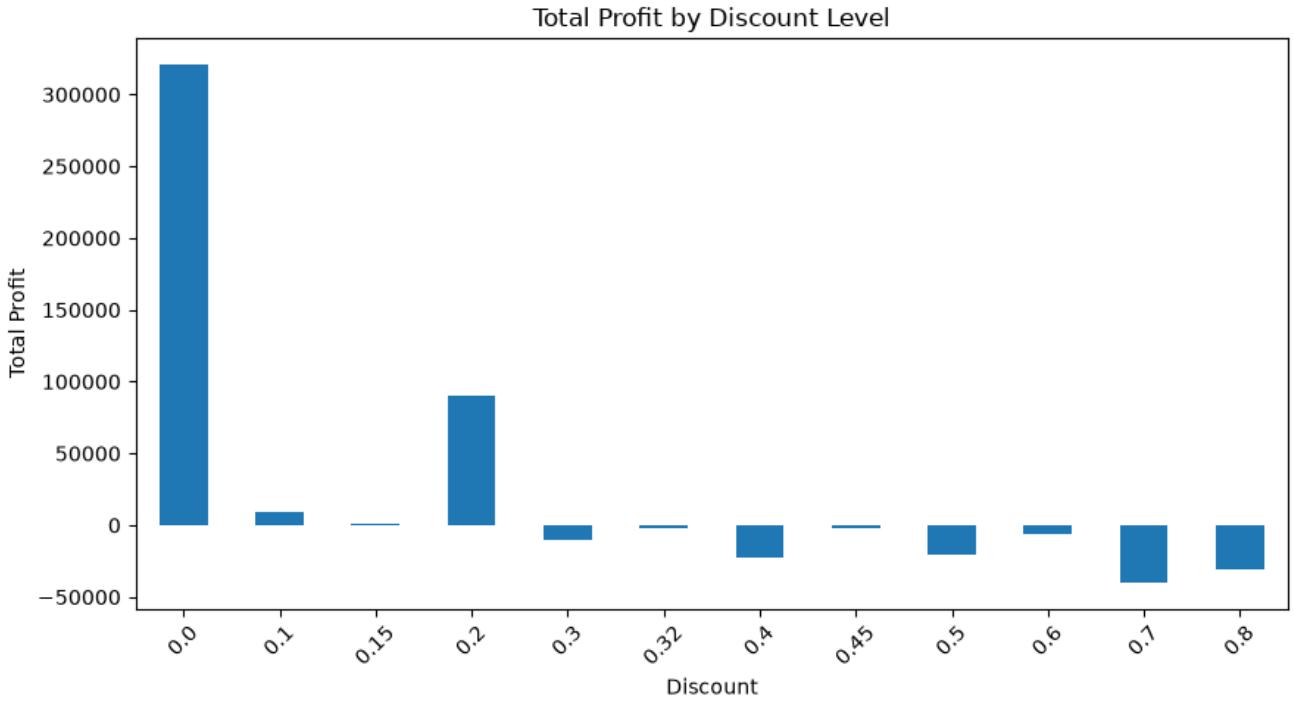
Interpretation: The category comparison shows that Furniture has a much lower profit margin than Technology and Office Supplies.

4. Profit by Sub-Category



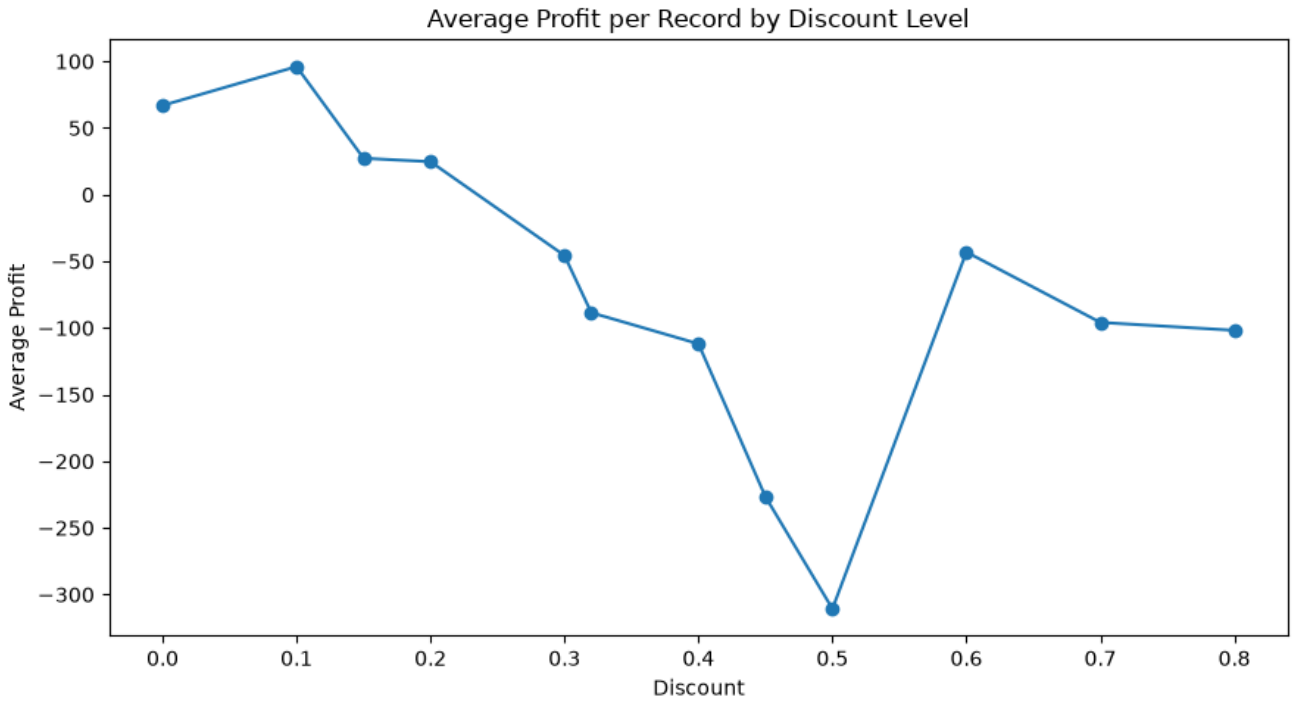
Interpretation: Sub-category profitability varies considerably. Tables is the most prominent negative-profit sub-category in the displayed analysis.

5. Total Profit by Discount Level



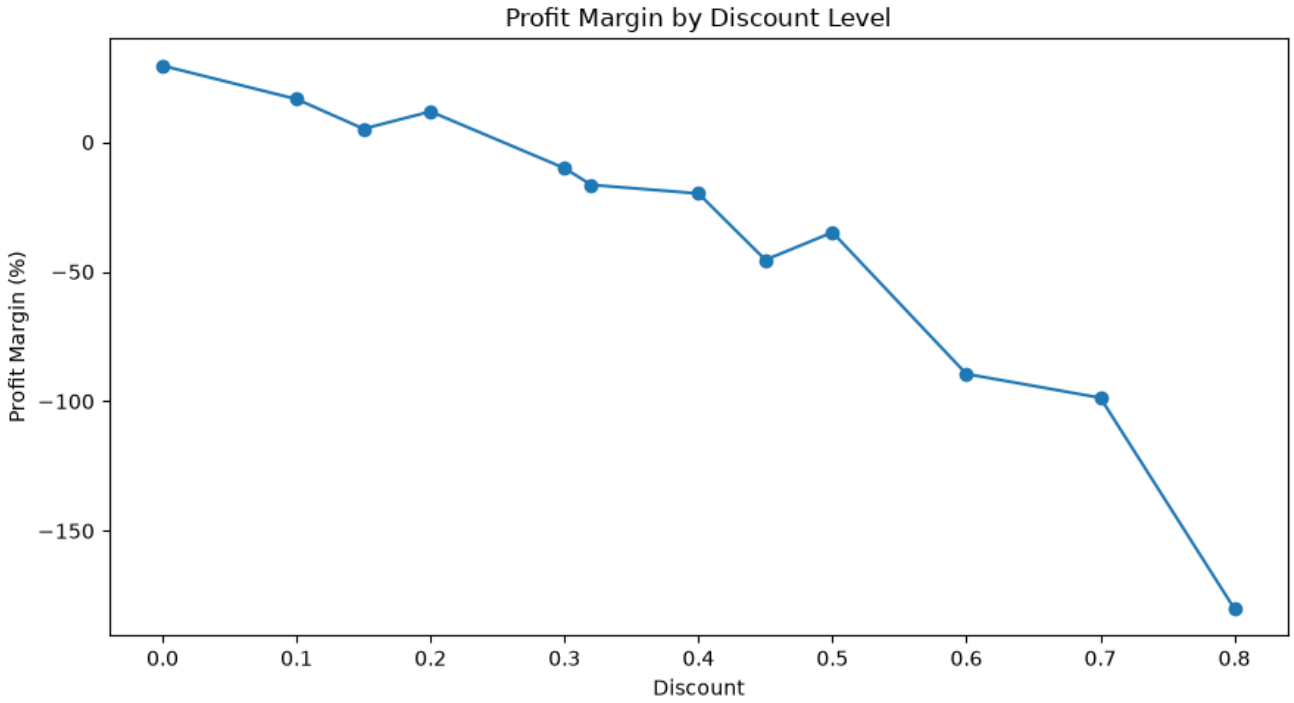
Interpretation: Profit varies substantially across discount levels. Some higher discount levels are associated with negative total profit.

6. Average Profit per Record by Discount



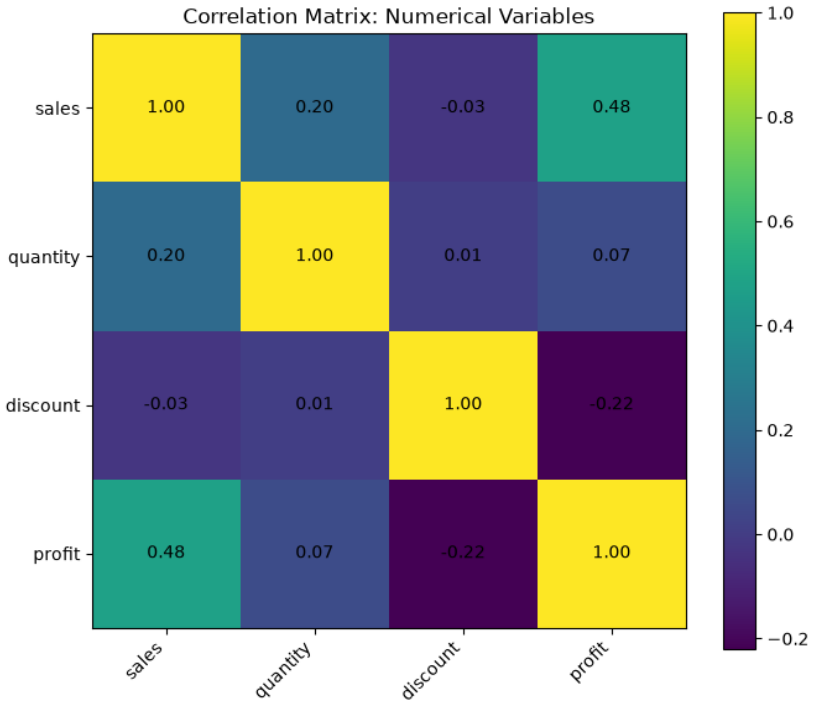
Interpretation: Average profit per record generally falls as discount levels become higher, with several discount levels showing negative average profit.

7. Profit Margin by Discount Level



Interpretation: Profit margin declines across higher discount levels in the displayed analysis, with the highest discount levels showing strongly negative margins.

8. Correlation Matrix



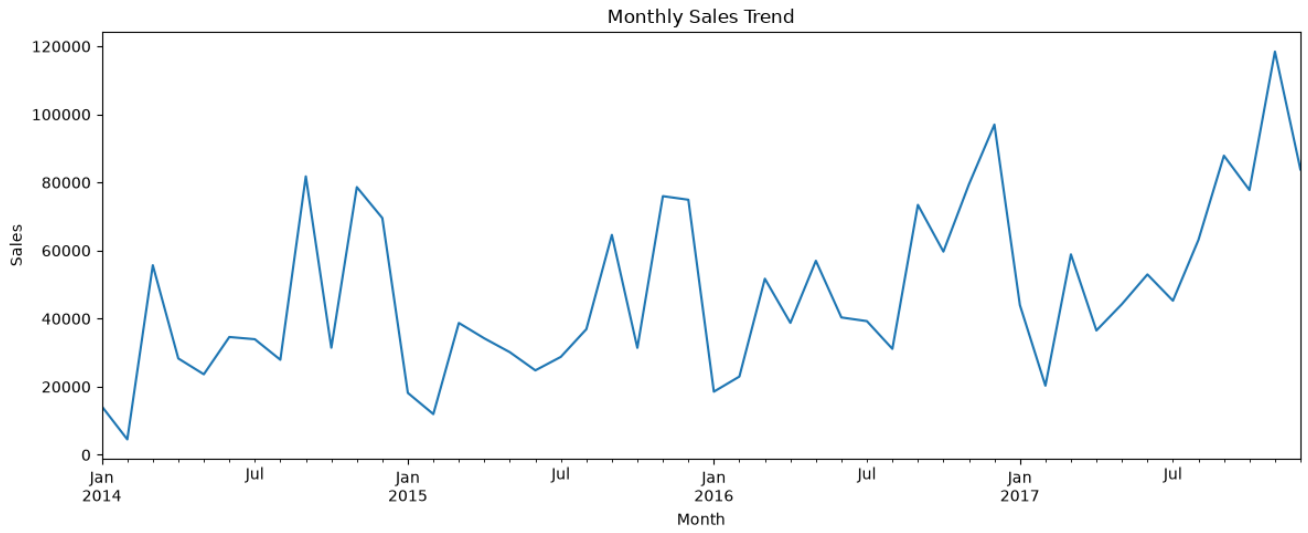
Interpretation: The matrix shows the linear relationships among sales, quantity, discount and profit. Correlation describes association and does not establish causation.

9. Sales vs Profit



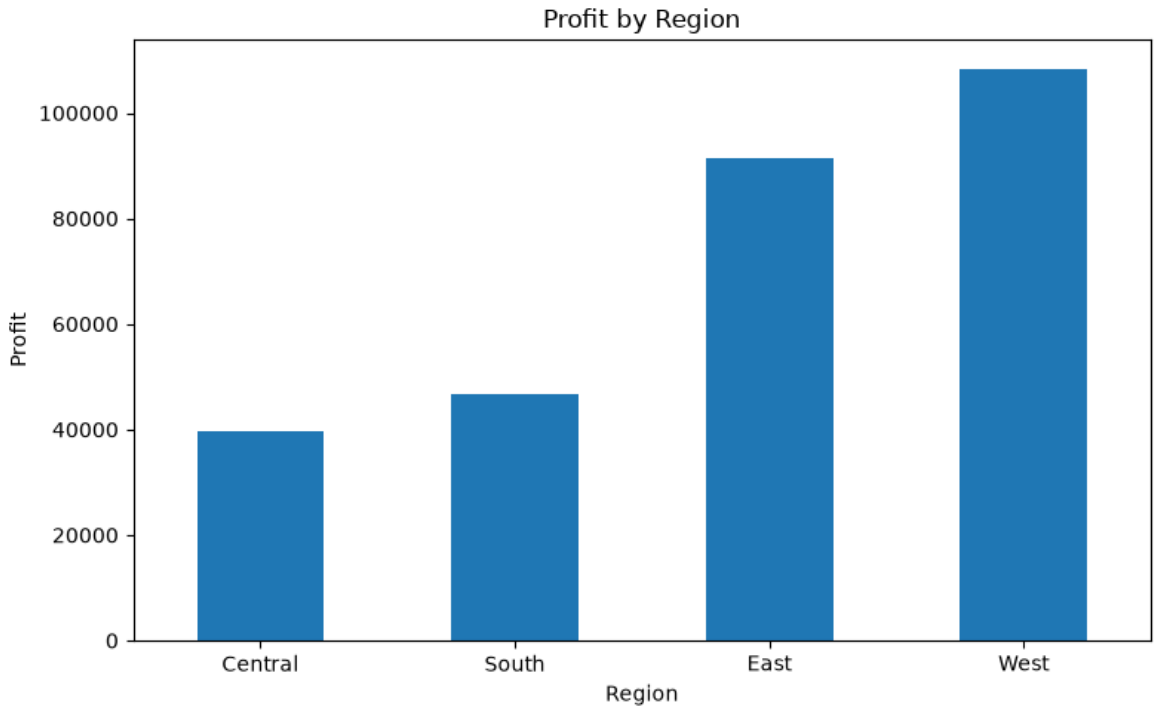
Interpretation: The scatter plot shows how profit varies across transaction sales values and highlights both positive and negative profit observations.

10. Monthly Sales Trend



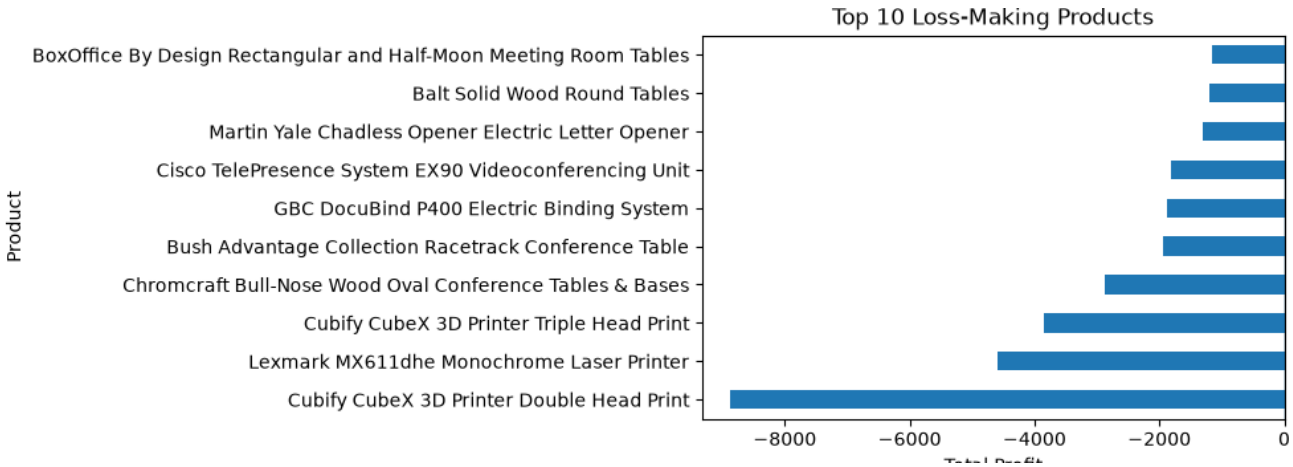
Interpretation: Monthly sales fluctuate over the analysed period, with noticeable variation between months and higher sales values toward the later part of the series.

11. Profit by Region



Interpretation: The regional comparison shows differences in total profit, with West and East having higher total profit than Central and South in the displayed analysis.

12. Top 10 Loss-Making Products



Interpretation: The chart highlights individual products with the lowest total profit, demonstrating why product-level analysis is useful when investigating loss areas.

Summary of Visual Findings

• **Category performance:** Sales and profit are not distributed equally across categories. Technology shows strong performance, while Furniture has a much lower profit margin.

• **Discount and profitability:** The visualisations show a clear association between higher discount levels and weaker profitability in parts of the dataset. This should be investigated further rather than interpreted as proof that discounts alone caused the losses.

• **Sub-category performance:** Tables stands out as an important loss-making sub-category and therefore warrants deeper product, geographic and discount analysis.

• **Regional performance:** Total profit differs across regions, providing another dimension for investigating where loss-making transactions are concentrated.

• **Product-level analysis:** The top-loss-products chart demonstrates that a small number of products can materially affect overall sub-category profitability.

• **Time trends:** Monthly sales vary considerably over time, making time-based analysis useful for understanding changes in business performance.

Interpretation note: These charts describe patterns in the Superstore dataset. Relationships shown by the visualisations should not automatically be interpreted as causal relationships. Further analysis would be required before concluding why a particular pattern occurs.